Example 3.30

(a) Solve the equation: $x - 7 = 6$

Solution:

$x - 7 = 6$ (Given)

$x - 7 + 7 = 6 + 7$ (add 7 on both sides)

$x = 13$

(b) Solve the equation: $3x = 51$

Solution:

$3x = 51$ (Given)

$3 \times x = 51$

$\frac{3 \times x}{3} = \frac{51}{3}$ (÷ 3 on both sides)

$x = 17$

(ii) Transposition method

The shifting of a number from one side of an equation to other is called transposition.

For the above example, (a) doing addition of 7 on both sides is the same as changing the number –7 on the left hand side to its additive inverse +7 and add it on the right hand side.

$x - 7 = 6$

$x = 6 + 7$

$x = 13$

likewise, (b) doing division by 3 on both sides is the same as changing the number 3 on the LHS to its reciprocal $\frac{1}{3}$ and multiply it on the RHS and vice-versa.

Think

Can you get more than one solution for a linear equation?

For Example (i) $6x = 12$

$x = \frac{12}{6}$

$x = 2$

(ii) $\frac{y}{7} = 10$

$y = 10 \times 7$

$y = 70$

Note

While rearranging the given linear equation, group the like terms on one side of the equality sign, and then do the basic arithmetic operations according to the signs that occur in the expression.

Example 3.31

Solve $2x + 5 = 9$

Solution:

$2x + 5 = 9$

$2x = 9 - 5$

$2x = 4$

$x = \frac{\cancel{4}^{2}}{\cancel{2}} = 2$

Example 3.32

Solve $\frac{4y}{3} - 7 = \frac{2y}{5}$

Solution:

(Rearranging the like terms)

$\frac{4y}{3} - \frac{2y}{5} = 7$

Try these

1. Solve for 'x' and 'y'

(i) $2x = 10$ (ii) $3 + x = 5$

(iii) $x - 6 = 10$ (iv) $3x + 5 = 2$

(v) $\frac{2x}{7} = 3$ (vi) $-2 = 4m - 6$

(vii) $4(3x - 1) = 80$ (viii) $3x - 8 = 7 - 2x$

(ix) $7 - y = 3(5 - y)$ (x) $4(1 - 2y) - 2(3 - y) = 0$

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